Keypad Based Product Counting and Automatic Batch Packing System with Station Sensing using LPC2129

ARM7 based embedded system | Two IR sensors, keypad, LCD and DC motor conveyor

1. Abstract

An LPC2129 (ARM7) microcontroller, two IR sensors and a keypad count products on a conveyor belt driven by a DC motor. The operator sets the batch quantity on the keypad and presses start. When the batch is complete the motor stops and the LCD shows the final count. If a counted product never reaches the packing station, the system reports a belt fault.

2
IR sensors: count and station check

1
Keypad to set batch quantity

0
Manual counting errors

2. Objectives

1. **Automate counting** — Count products on the belt with an external interrupt, not by hand.
2. **Set batch size** — Let the operator enter the batch quantity through the keypad.
3. **Detect belt faults** — Spot a missing, fallen or blocked product at the packing station.
4. **Indicate status** — Show count on the LCD, with buzzer and red / green LEDs.

3. Hardware Components

Component	Role
LPC2129	ARM7 main controller
4x4 Keypad	Batch quantity and START
IR Sensor 1	Counts at belt start
IR Sensor 2	Checks packing station
16x2 LCD	Count and fault message
DC Motor	Drives the conveyor
Motor Driver	L293D or relay stage
Buzzer + LEDs	Red running, green done

4. Block Diagram

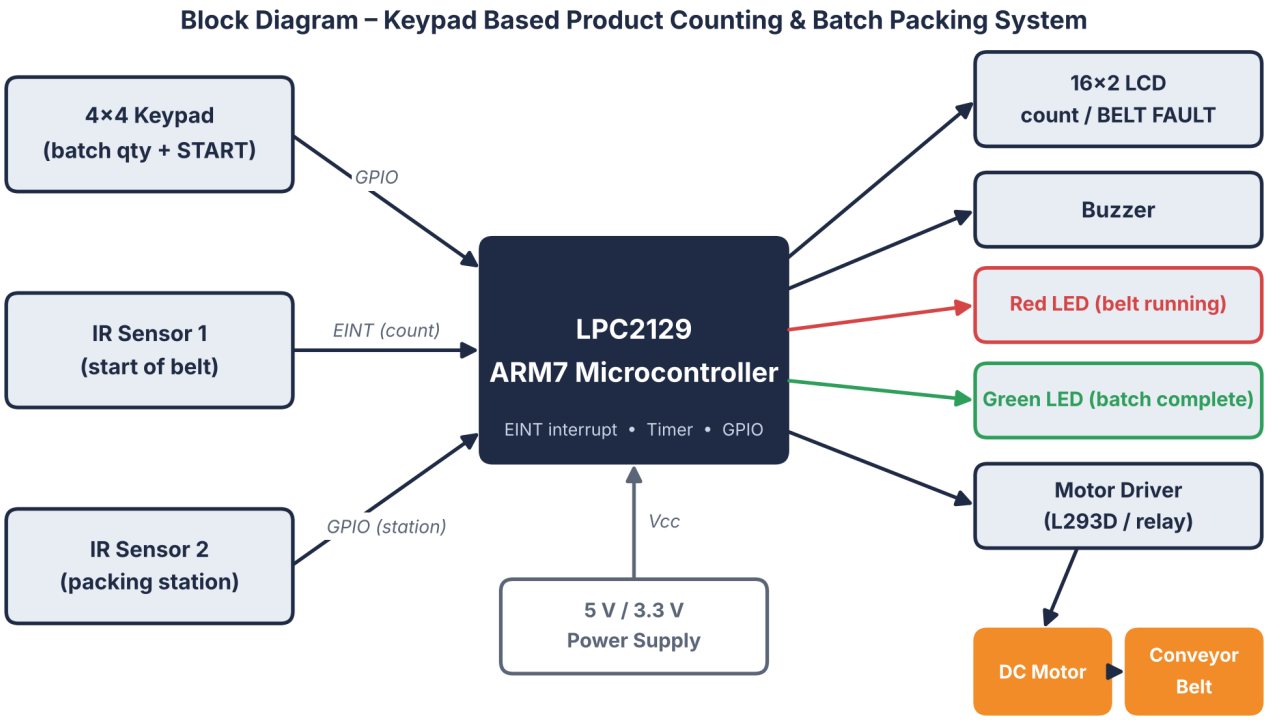


Figure 1. Block diagram of the system

5. Working Principle

Step 1. Operator enters the batch quantity on the keypad and presses START.

Step 2. Motor starts the conveyor and the red LED glows.

Step 3. IR sensor 1 at belt start raises an interrupt and the count increases.

Step 4. IR sensor 2 at the packing station confirms each product arrived.

Step 5. No detection within the set time: motor stops, LCD shows BELT FAULT.

Step 6. Count equals batch: motor stops, buzzer beeps, green LED glows, LCD shows final count.

6. Flowchart

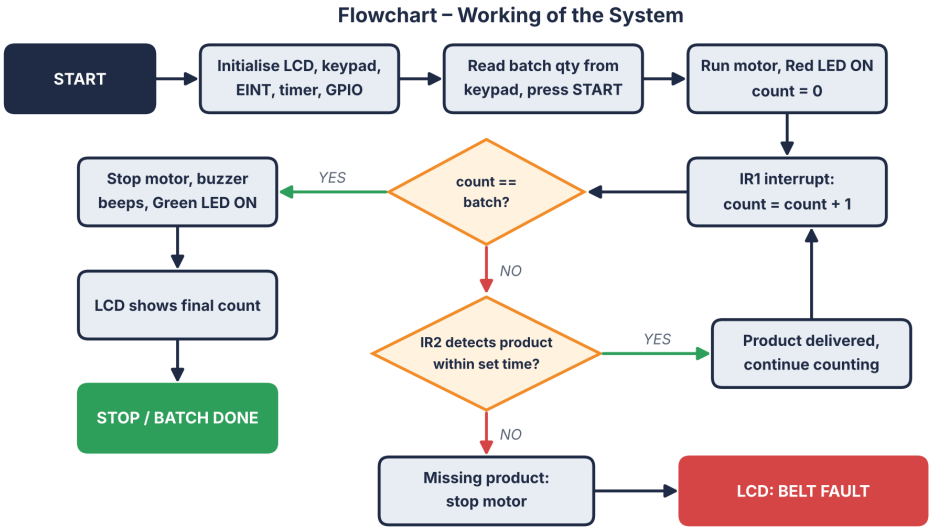


Figure 2. Flowchart – working of the system

7. System Output Indications

Condition	Motor	LCD	LED	Buzzer
Belt running	ON	Live count	Red	Off
Batch complete	OFF	Final count	Green	Beeps
Missing product	OFF	BELT FAULT	Red off	Off

Fault logic: IR1 counted a product, but IR2 did not see it at the packing station within the set time.

8. Advantages and Applications

Advantages	Applications
• Removes manual counting errors	• Packaging lines
• Detects fallen or blocked products	• Bottling and carton filling
• Low cost, simple ARM7 design	• Production line batch counting
• Clear LCD, LED and buzzer feedback	• Warehouse conveyors

9. Conclusion

• The LPC2129 based system counts products accurately using an interrupt driven IR sensor.

• A second IR sensor at the packing station catches missing products and raises a BELT FAULT.

• It lowers manual counting errors and suits packaging and production lines.